IS222: Database Management Systems

Assignment 1

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1.0 Introduction

1.1 Introduction to the Business Process

In a normal business environment, any operation that requires a series of interconnected tasks are know as business processes. SuvaTech Solutions is a leading Fijian IT consultancy firm that provides specialized services across multiple sectors: Government, Enterprise, and SMEs.
The core business process of this firm are managing client contracts through service-level agreements (SLA), project scopes, renewal terms and compliancce requirements. Moreover, the firm assigns consultants to projects depending on their expertise level and secuirty clearance. Additionally, to allow consultants to work on the project, consultants require training sessions and certification. Futhermore, the organization manages client subscriptions inregards to cloud services and software licenses.
To manage all these processes in the most efficient way it is therefore neccessay to utilize a database. A database is a collection of data that is stored and can be manipulated to find important queries, important information and records

1.2 Role of the Database in the Business Process

The database plays a central role in:

- Storing data related to clients, contracts, consultants, projects and training
- Creating relationships between the different entites and enforcing referencial integrity between each table.
- Having cascading updating and deletion whenever a entry is changed or deleted.
- Acting as a way to quickly access the data and automate tasks such as report generation for number of contracts related to a specific client.

1.3 Advantages of Using a Database

- Data management: Allows for better data accuracy and reduces data redundancy
- Data retrieval: Enables to quickly retrieve data through queries and reports
- Automation: Makes easy generation of forms and reports to make analyzing data easier.
- Hard-data: The structured and accurate data will aid in making informed decisions in business decision making.
- Operation streamlining: Removes need for tabular data entry through use of forms and has a centralized system from which data can be retrieved.

2.0 Business Rules

1. A client can have zero or many contracts whereas each contracts can have only one and one client.

    - Client can choose to have none to multiple contracts. However, every contract must be tied to a specific client.

2. Each business divison can handle zero to many contracts whereas each contract can belong to only one and one business division.

    - Contracts are handled by different business divisions, and each contract falls under the responsibility of a specific division.

3. A contract can have one to many projects to be completed whereas each project can only belong to one and only one contract.

    - Contracts can have multiple projects within itself that make up the contract whereas each project is just paart of the whole contract.

4. Each project can have one to many consultants assigned to it whereas each consultant can be assigned to zero to many projects.

    - Projects can have multiple consultants working on it, while it is also possible that a consultant is working on zero projects or doesn't meet the requirements to work on it.

5. Each consultant can train for zero to many training sessions whereas each training session can be attended by zero to many consultants.

    - Consultants may undergo multiple training sessions, and sessions may have multiple attendees.

6. Each training session can contribute to one and only one certificate whereas each certificate will have one to many training sessions.

    - Certificates are achieved through one or more sessions. One session counts for One certification.

7. Each consultant can have zero to many certifcations whereas each certifcate can be given to zero to many consultants.

    - There can be certifications which no consultants have while a consultant can also have none certifications or many.

8. Each client can have zero to many subsriptions which they can upgrade, downgrade and renew whereas each subscription applies to only one client.

    - A subscription purchase will be down only 1 client whereas each client can have multiple or none subscriptions purchased.

9. Each subscription can consiste of only one and one pricing tier whereas each pricing tier can have zero to many subscriptions.

    - Each subscription purchase can be at only 1 price tier while each pricing tier can have no one having purchased a subscription for it.

10. Each pricing tier will be for one and only one cloud service whereas each cloud service can have one to many pricing tiers.

    - Cloud services offer different levels of features at different prices. But the pricing tier price will be for only one particular cloud service

11. Each pricing tier will be for one and only one software license whereas each software license can have one to many pricing tiers.

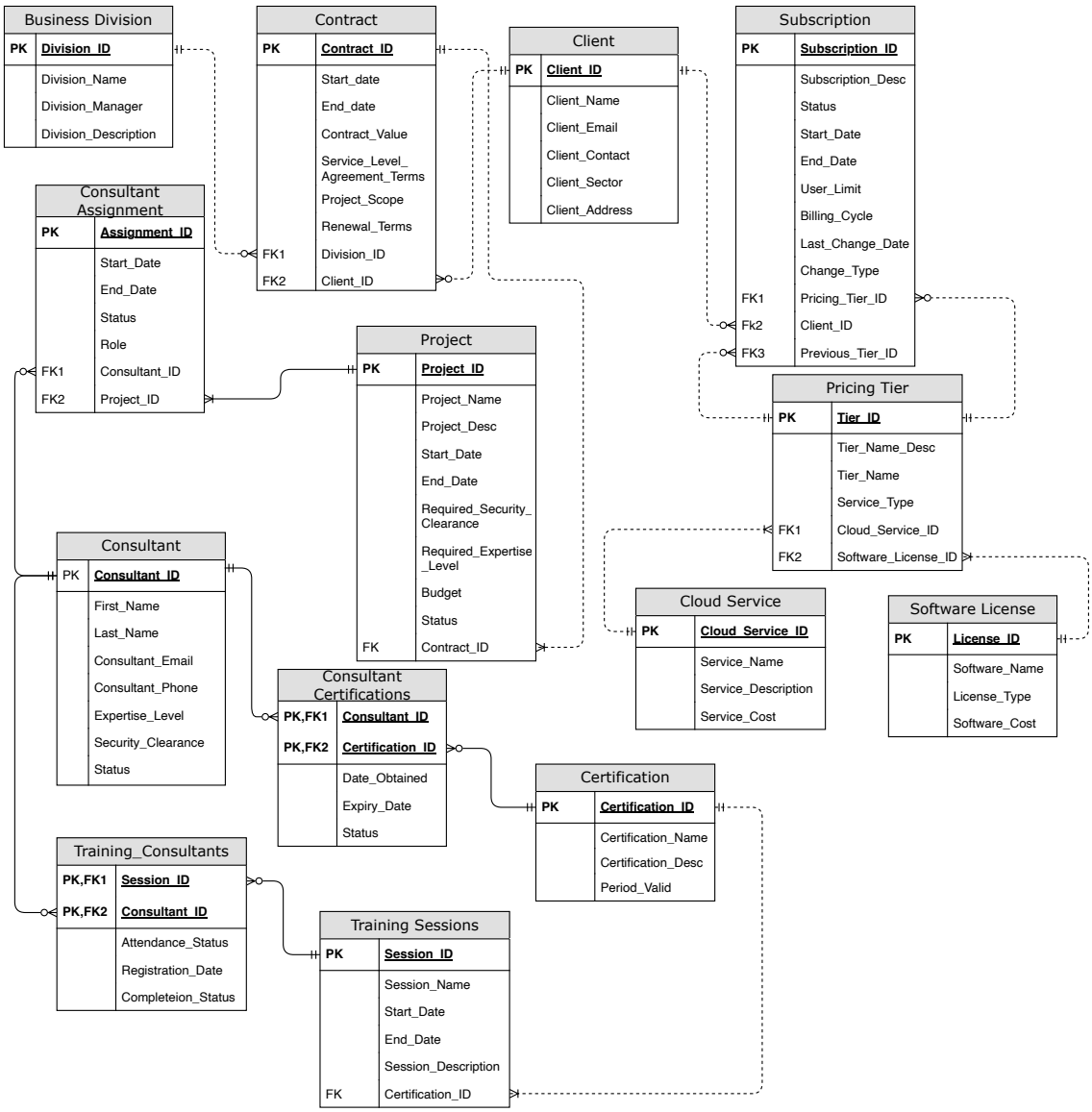
    - Software license offer different levels of features at different prices. But the pricing tier price will be for only one software license.

2.1 Assumptions

1. Each consultant can have one and only one expertise level whereras each expertise level can be associated with zero to many consultants.
    a. Reason
        i. Many projects require consultants of different expertise level; Junior, Senior and Principal. Consultants will assigned to a project depending on whether they meet the requirement.
2. Each consultant can have one and only one security whereras each security clearance can be associated with zero to many consultants.
    a. Reason:
        i. Many projects may require different levels of security; confidential to top secret. Consultants will assigned to a project depending on whether they meet the requirement.
3. Each project requires a consultant to have a one to many certifications to be able to work on a project, whereas each certificate enables a consultant to qualify to work on zero to many projects.
    a. Reason:

i. Projects require the consultant to have specific certifcations to be able to work on a particular project. Expiry of the certification may stop a consultant from working on a particular project.

3.0 Entity Relationship Diagram



4.0 Data Dictionary - Consultant Management System

Table Name	Attribute Name	Contents	Type	Format	Required	PK or FK	FK Referenced Table
Client	Client_ID	Client identifier	Long Integer (4)	9999	Y	PK	-
Client	Client_Name	Name of client	VARCHAR(35)	-	Y	-	-
Client	Client_Email	Client's email address	VARCHAR(35)	-	Y	-	-
Client	Client_Contact	Client's contact number	CHAR(8)	999-9999	Y	-	-
Client	Client_Sector	Client's industry sector (Health,	VARCHAR(15)	-	Y	-	-
Client	Client_Address	Client's Physical address	VARCHAR(35)	-	Y	-	-
Business Division	Division_ID	Division identifier	Long Integer (4)	9999	Y	PK	-
Business Division	Division_Name	Name of business division	VARCHAR(35)	-	Y	-	-

Business Division	Division_Manager	Manager of division	VARCHAR(35)	-	Y	-	-
Business Division	Division_Description	Description of division	VARCHAR(35)	-	N	-	-
Contract	Contract_ID	Contract identifier	CHAR(4)	9999	Y	PK	-
Contract	Start_date	Contract start date	Date(8)	dd-mm-yy	Y	-	-
Contract	End_date	Contract end date	Date(8)	dd-mm-yy	Y	-	-
Contract	Contract_Value	Value of contract	Number(8)	9,999,999.99	Y	-	-
Contract	Service_Level_Agreement_Terms	SLA terms	VARCHAR(35)	-	Y	-	-
Contract	Project_Scope	Scope of the project	VARCHAR(35)	-	Y	-	-
Contract	Renewal_Terms	Terms for renewal	VARCHAR(35)	-	Y	-	-
Contract	Division_ID	Division identifier	Long Integer (4)	9999	Y	FK	Business Division
Contract	Client_ID	Client identifier	Long Integer (4)	9999	Y	FK	Client
Subscription	Subscription_ID	Subscription identifier	CHAR(4)	9999	Y	PK	-
Subscription	Subscription_Desc	Description of	VARCHAR(35)	-	Y	-	-

		subscription					
Subscription	Status	Current status	VARCHAR(15)	-	Y	-	-
Subscription	Start_Date	Start date	Date(8)	dd-mm-yy	Y	-	-
Subscription	End_Date	End date	Date(8)	dd-mm-yy	Y	-	-
Subscription	User_Limit	User limit	Long Integer (4)	9999	Y	-	-
Subscription	Billing_Cycle	Billing cycle period	VARCHAR(15)	-	Y	-	-
Subscription	Last_Change_Date	Date of last change	Date(8)	dd-mm-yy	Y	-	-
Subscription	Change_Type	Type of change	VARCHAR(15)	-	Y	-	-
Subscription	Pricing_Tier_ID	Pricing tier identifier	CHAR(4)	9999	Y	FK	Pricing Tier
Subscription	Client_ID	Client identifier	Long Integer (4)	9999	Y	FK	Client
Subscription	Previous_Tier_ID	Previous tier identifier	CHAR(4)	9999	N	FK	Pricing Tier
Project	Project_ID	Project identifier	CHAR(4)	9999	Y	PK	-
Project	Project_Name	Name of project	VARCHAR(35)	-	Y	-	-

Project	Project_Desc	Project description	VARCHAR(35)	-	Y	-	-
Project	Start_Date	Start date	Date(8)	dd-mm-yy	Y	-	-
Project	End_Date	End date	Date(8)	dd-mm-yy	Y	-	-
Project	Required_Security_Clearance	Required security clearance level	VARCHAR(15)	-	Y	-	-
Project	Required_Expertise_Level	Required expertise level	VARCHAR(15)	-	Y	-	-
Project	Budget	Project budget	Number(8)	9,999,999.99	Y	-	-
Project	Status	Current status	VARCHAR(15)	-	Y	-	-
Project	Contract_ID	Contract identifier	CHAR(4)	9999	Y	FK	Contract
Certification	Certification_ID	Certification identifier	CHAR(4)	9999	Y	PK	-
Certification	Certification_Name	Name of certification	VARCHAR(35)	-	Y	-	-
Certification	Certification_Desc	Description of certification	VARCHAR(35)	-	Y	-	-

Certification	Period_Valid	Period of validity	VARCHAR(15)	-	Y	-	-
Consultant	Consultant_ID	Consultant identifier	CHAR(4)	9999	Y	PK	-
Consultant	First_Name	Consultant's first name	VARCHAR(15)	-	Y	-	-
Consultant	Last_Name	Consultant's last name	VARCHAR(15)	-	Y	-	-
Consultant	Consultant_Email	Consultant's email address	VARCHAR(35)	-	Y	-	-
Consultant	Consultant_Phone	Consultant's phone number	CHAR(8)	999-9999	Y	-	-
Consultant	Expertise_Level	Consultant's level of expertise	VARCHAR(15)	-	Y	-	-
Consultant	Security_Clearance	Consultant's security clearance level	VARCHAR(15)	-	Y	-	-
Consultant	Status	Consultant's Current status	VARCHAR(15)	-	Y	-	-

Cloud Service	Cloud_Service_ID	Cloud service identifier	CHAR(4)	9999	Y	PK	-
Cloud Service	Service_Name	Name of service	VARCHAR(35)	-	Y	-	-
Cloud Service	Service_Description	Description of service	VARCHAR(35)	-	Y	-	-
Cloud Service	Service_Cost	Cost of service	Number(8)	9,999,999.99	Y	-	-
Software License	License_ID	License identifier	CHAR(4)	9999	Y	PK	-
Software License	Software_Name	Name of software	VARCHAR(35)	-	Y	-	-
Software License	License_Desc	Description of software	VARCHAR(15)	-	Y	-	-
Software License	Software_Cost	Cost of software	Number(8)	9,999,999.99	Y	-	-
Training Sessions	Session_ID	Session identifier	CHAR(4)	9999	Y	PK	-
Training Sessions	Session_Name	Name of session	VARCHAR(35)	-	Y	-	-
Training Sessions	Start_Date	Start date	Date(8)	dd-mm-yy	Y	-	-
Training Sessions	End_Date	End date	Date(8)	dd-mm-yy	Y	-	-
Training Sessions	Session_Description	Description of session	VARCHAR(35)	-	Y	-	-

Training Sessions	Certification_ID	Certification identifier	CHAR(4)	9999	Y	FK	Certification
Pricing Tier	Tier_ID	Tier identifier	CHAR(4)	9999	Y	PK	-
Pricing Tier	Tier_Name_Desc	Tier name description	VARCHAR(35)	-	Y	-	-
Pricing Tier	Tier_Name	Name of tier	VARCHAR(15)	-	Y	-	-
Pricing Tier	Service_Type	Type of service	VARCHAR(15)	-	Y	-	-
Pricing Tier	Cloud_Service_ID	Cloud service identifier	CHAR(4)	9999	N	FK	Cloud Service
Pricing Tier	Software_License_ID	Software license identifier	CHAR(4)	9999	N	FK	Software License
Consultant Assignment	Assignment_ID	Assignment identifier	CHAR(4)	9999	Y	PK	-
Consultant Assignment	Start_Date	Start date	Date(8)	dd-mm-yy	Y	-	-
Consultant Assignment	End_Date	End date	Date(8)	dd-mm-yy	Y	-	-
Consultant Assignment	Status	Current status	VARCHAR(15)	-	Y	-	-

Consultant Assignment	Role	Role in assignment	VARCHAR(15)	-	Y	-	-
Consultant Assignment	Consultant_ID	Consultant identifier	CHAR(4)	9999	Y	FK	Consultant
Consultant Assignment	Project_ID	Project identifier	CHAR(4)	9999	Y	FK	Project
Training_Consultants	Session_ID	Session identifier	CHAR(4)	9999	Y	PK/FK	Training Sessions
Training_Consultants	Consultant_ID	Consultant identifier	CHAR(4)	9999	Y	PK/FK	Consultant
Training_Consultants	Attendance_Status	Status of attendance	VARCHAR(15)	-	Y	-	-
Training_Consultants	Registration_Date	Date of registration	Date(8)	dd-mm-yy	Y	-	-
Training_Consultants	Completion_Status	Status of completion	VARCHAR(15)	-	Y	-	-
Consultant Certifications	Consultant_ID	Consultant identifier	CHAR(4)	9999	Y	PK/FK	Consultant
Consultant Certifications	Certification_ID	Certification identifier	CHAR(4)	9999	Y	PK/FK	Certification
Consultant Certifications	Date_Obtained	Date certification obtained	Date(8)	dd-mm-yy	Y	-	-

Consultant Certifications	Expiry_Date	Expiration date	Date(8)	dd-mm-yy	Y	-	-
Consultant Certifications	Status	Current status	VARCHAR(15)	-	Y	-	-

5.0 Conclusion

To conclude, the main achievement made throughout this assignment was being able to design a database system for SuvaTech Solutions. The processes involved analyzing the sample text, abstracting the business rules from it and creating an Entity-Relationship Diagram (ERD) using the information. A data dictionary was also created to find the appropriate data type for each attribute.

Furthermore, some of the resources used to complete this assignment include the course textbook, lecture slides and tutorials as well as practical implementation of the database system into Microsoft Access.

Using these resources I was able to acquire a comprehensive understanding on the importance of databases and how to decipher business rules and make an ERD.

Additionally, challenges encountered during the assignment included deciphering the business rules from the scenario and ensuring the ERD is able to accurately depict all of the rules. Moreover, another challenge include finding what would be strong and weak relationships and the type of key each entity would have.